TRIPATHI STUDY ZONE

JEE MAIN | NEET | AIIMS | KVPY | BITSAT

Mock Test 4

Time : 3 hrs. **Max. Marks : 300**

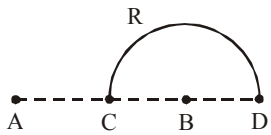
INSTRUCTIONS

1. This test will be a 3 hours Test.
2. This test consists of Physics, Chemistry and Mathematics questions with equal weightage of 100 marks.
3. Each question is of 4 marks.
4. There are three parts in the question paper consisting of Physics (Q.no.1 to 30), Chemistry (Q.no.31 to 60) and Mathematics (Q. no.61 to 90). Each part is divided into two sections, Section A consists of 20 multiple choice questions & Section B consists of 10 Numerical value answer Questions.
   In Section B, candidates have to attempt **only 5 questions out of 10**.
5. There will be only one correct choice in the given four choices in Section A. For each question 4 marks will be awarded for correct choice, 1 mark will be deducted for incorrect choice and zero mark will be awarded for unattempted question. For Section B 4 marks will be awarded for correct answer and zero for marked for each review / unattempted/incorrect answer.
6. Any textual, printed or written material, mobile phones, calculator etc. is not allowed for the students appearing for the test.
7. All calculations / written work should be done in the rough sheet provided.

PHYSICS

Section - A

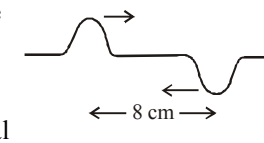
1. Charges +q and –q are placed at points A and B respectively which are a distance 2L apart, C is the midpoint between A and B. The work done in moving a charge +Q along the semicircle CRD is



(1) $\frac{qQ}{2\pi\varepsilon_0 L}$ (2) $\frac{qQ}{6\pi\varepsilon_0 L}$

(3) $-\frac{qQ}{6\pi\varepsilon_0 L}$ (4) $\frac{qQ}{4\pi\varepsilon_0 L}$

2. Two pulses in a stretched string whose centres are initially 8 cm apart are moving towards each other as shown in the figure. The speed of each pulse is 2 cm/s. After 2 seconds, the total energy of the pulses will be



(1) Zero
(2) Purely kinetic
(3) Purely potential
(4) Partly kinetic and partly potential

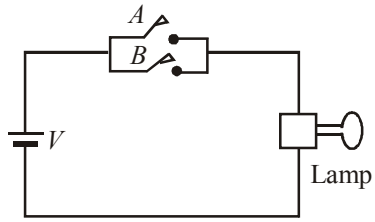
Space for Rough Work

3. A metallic wire of density d is lying horizontal on the surface of water. The maximum length of wire so that it may not sink will be

(1) $\sqrt{\frac{2Tg}{\pi d}}$ (2) $\sqrt{\frac{2\pi T}{dg}}$

(3) $\sqrt{\frac{2T}{\pi dg}}$ (4) any length

4. Which logic gate with inputs A and B performs the same operation as that performed by the following circuit?



(1) NAND gate (2) OR gate
(3) NOR gate (4) AND gate

5. The specific heat capacity of a metal at low temperature (T) is given as $C_p(\text{kJK}^{-1}\text{kg}^{-1}) = 32\left(\frac{T}{400}\right)^3$. A 100 g vessel of this metal is to be cooled from 20 K to 4 K by a special refrigerator operating at room temperature (27°C). The amount of work required to cool in vessel is

(1) equal to 0.002 kJ
(2) greater than 0.148 kJ
(3) between 0.148 kJ and 0.028 kJ
(4) less than 0.028 kJ

6. A pendulum made of a uniform wire of cross sectional area A has time period T. When an additional mass M is added to its bob, the time period changes to T_M. If the Young's modulus of the material of the wire is Y then $\frac{1}{Y}$ is equal to: (*g = gravitational acceleration*)

(1) $\left[1-\left(\frac{T_M}{T}\right)^2\right]\frac{A}{Mg}$ (2) $\left[1-\left(\frac{T}{T_M}\right)^2\right]\frac{A}{Mg}$

(3) $\left[\left(\frac{T_M}{T}\right)^2-1\right]\frac{A}{Mg}$ (4) $\left[\left(\frac{T_M}{T}\right)^2-1\right]\frac{Mg}{A}$

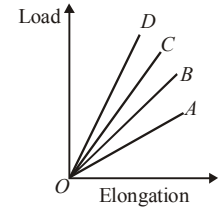
7. A bucket full of hot water is kept in a room and it cools from 75°C to 70°C in T_1 minutes, from 70°C to 65°C in T_2 minutes and from 65°C to 60°C in T_3 minutes. Then

(1) $T_1 = T_2 = T_3$ (2) $T_1 < T_2 < T_3$
(3) $T_1 > T_2 > T_3$ (4) $T_1 < T_3 < T_2$

8. The length of an elastic string is x when the tension is 5N. Its length is y when the tension is 7N. What will be its length, when the tension is 9N?

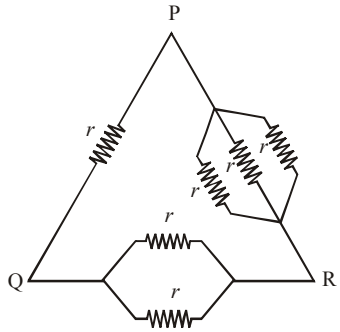
(1) $2y + x$ (2) $2y - x$
(3) $7x - 5y$ (4) $7x + 5y$

9. The load versus elongation graphs for four wires of same length and made of the same material are shown in the figure. The thinnest wire is represented by the line



(1) *OA* (2) *OC*
(3) *OD* (4) *OB*

10. Six equal resistances are connected between points P, Q and R as shown in figure. Then net resistance will be maximum between :



(1) P and R (2) P and Q

(3) Q and R (4) Any two points

11. Electrons are accelerated through a potential difference V and protons are accelerated through a potential difference 4 V. The de-Broglie wavelengths are λ_e and λ_p for electrons and protons respectively. The ratio of $\frac{\lambda_e}{\lambda_p}$ is given by : (given m_e is mass of electron and m_p is mass of proton).

(1) $\frac{\lambda_e}{\lambda_p} = \sqrt{\frac{m_p}{m_e}}$ (2) $\frac{\lambda_e}{\lambda_p} = \sqrt{\frac{m_e}{m_p}}$

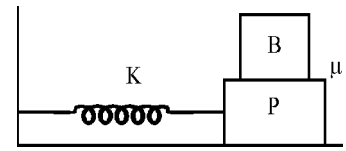
(3) $\frac{\lambda_e}{\lambda_p} = \frac{1}{2}\sqrt{\frac{m_e}{m_p}}$ (4) $\frac{\lambda_e}{\lambda_p} = 2\sqrt{\frac{m_p}{m_e}}$

12. A straight section PO of a circuit lies along the x-axis from x = –a/2 to x = +a/2, and carries a steady current 'I'. The magnitude of magnetic field due to the section PO at a point to y = + a will be

(1) proportional to a (2) proportional to a^2

(3) proportional to 1/a (4) equal to zero

13. A flat plate P of mass 'M' executes SHM in a horizontal plane by sliding over a frictionless surface with a frequency V. A block 'B' of mass 'm' rests on the plate as shown in figure. Coefficient of friction between the surface of B and P is μ. What is the maximum amplitude of oscillation that the plate block system can have if the block B is not to slip on the plate :



(1) $\frac{\mu g}{4\pi^2 V^2}$ (2) $\frac{\mu g}{4\pi^2 V}$

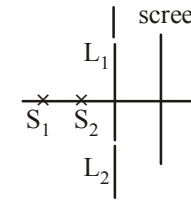
(3) $\frac{\mu}{4\pi^2 V^2 g}$ (4) $\frac{\mu g}{2\pi^2 V^2}$

14. A glass slab has the left half of refractive index n_1, and the right half of $n_2 = 3n_1$. The effective refractive index of the whole slab is

(1) $\frac{n_1}{2}$ (2) 2n

(3) $\frac{3n_1}{2}$ (4) $\frac{2n_1}{3}$

15. In the arrangement shown L_1, L_2 are slits and S_1 and S_2 are two independent sources. On the screen, interference fringes



(1) will not be there

(2) will not be there if the intensity of light reaching the screen from S_1 and S_2 are equal.

(3) will be there under all circumstances

(4) we will have only the central fringe

16. For good demodulation of AM signal of carrier frequency f, the value of RC should be

(1) $RC = \frac{1}{f}$ (2) $RC < \frac{1}{f}$

(3) $RC \geq \frac{1}{f}$ (4) $RC >> \frac{1}{f}$

17. The radioactivity of a sample is R_1 at a time T_1 and R_2 at a time T_2. If the half life of the specimen is T, the number of atoms that have disintegrated in the time $(T_2 - T_1)$ is proportional to

(1) $(R_1T_1 - R_2T_2)$ (2) $(R_1 - R_2)$

(3) $(R_1 - R_2)/T$ (4) $(R_1 - R_2) \times T$

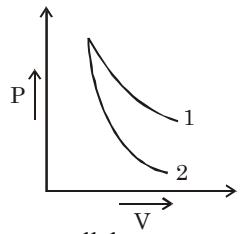
18. P–V plots for two gases during adiabatic processes are shown in the figure. Plots 1 and 2 should correspond respectively to

(1) He and Ar

(2) He and O_2

(3) O_2 and N_2

(4) O_2 and He



19. A magnetic needle lying parallel to a magnetic field requiers W units of work to turn it through 60^0. The torque needed to maintain the needle in this position will be

(1) $\sqrt{3}$W (2) W

(3) $\frac{\sqrt{3}}{2}$W (4) 2W

20. Two small equal point charges of magnitude q are suspended from a common point on the ceiling by insulating mass less strings of equal lengths. They come to equilibrium with each string making angle θ from the vertical. If the mass of each charge is m, then the electrostatic potential at the centre of line joining them will be $\left(\frac{1}{4\pi \in_0} = k\right)$.

(1) $2\sqrt{k\,mg\tan\theta}$ (2) $4\sqrt{k\,mg\tan\theta}$

(3) $4\sqrt{k\,mg/\tan\theta}$ (4) $\sqrt{k\,mg/\tan\theta}$

Section - B

21. A car is standing 200 m behind a bus, which is also at rest. The two start moving at the same instant but with different forward accelerations. The bus has acceleration 2 m/s^2 and the car has acceleration 4 m/s^2. The car will catch up the bus after a time of s.

22. A transformer is used to light a 140 W, 24 V bulb from a 240 V a.c. mains. The current in the main cable is 0.7 A. The efficiency of the transformer is _____%.

23. A rod of length L is placed on x – axis between x = 0 and x = L. The linear density i.e., mass per unit length denoted by ρ, of this rod, varies as, $\rho = a + bx$. The dimensions of b is $ML^{-2}T^x$. Find the value of x.

24. The surface charge density of a thin charged disc of radius R is σ. The value of the electric field at the centre of the disc is $\frac{\sigma}{2\in_0}$. With respect to the field at the centre, the electric field along the axis at a distance R from the centre of the disc is reduces by _____%.

25. A body is thrown vertically upwards from the surface of earth in such a way that it reaches upto a height equal to $10R_e$. The velocity imparted to the body will be km/s

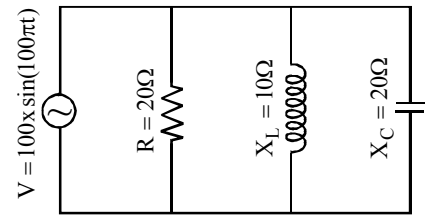
26. What is the ratio of the circumference of the first Bohr orbit for the electron in the hydrogen atom to the de Brogile wavelength of electrons having the same velocity as the electron in the first Bohr orbit of the hydrogen atom?

27. A projectile can have the same range 'R' for two angles of projection. If 'T_1' and 'T_2' to be time of flights in the two cases, then the product of the two time of flights is $\frac{xR}{g}$. Find the value of x.

28. The electric field associated with an e.m. wave in vacuum is given by $\vec{E} = \hat{i}\ 40 \cos(kz - 6 \times 10^8 t)$, where E, z and t are in volt/m, meter and seconds respectively. The value of wave vector k is m^{-1}.

29. Two points of a rod move with velocities 3v and v perpendicular to the rod and in the same direction, separated by a distance r. Then the angular velocity of the rod $\frac{xv}{r}$. Find the value of x.

30. In the given circuit, the current drawn from the source is _____A.

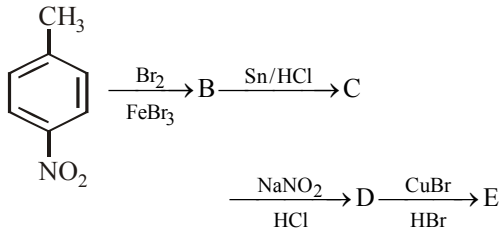


CHEMISTRY

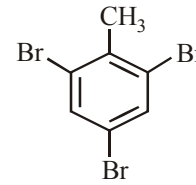
Section - A

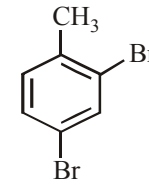
31. Which of the following structures does not contain any chiral C atom but represent the chirality in the structure.
(1) 2 – Ethyl – 3 – hexene
(2) 2, 3-Pentadiene
(3) 1,3 – Butadiene
(4) Pent – 3 – en – 1 – yne

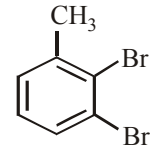
32. In a set of reactions *p*-nitrotoluene yielded a product E.

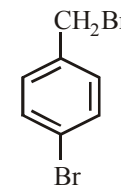


The product E would be:

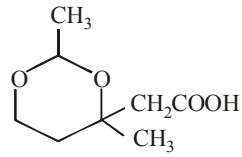
(1) 

(2) 

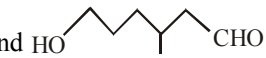
(3) 

(4) 

33. Compound A is formed by the interaction of

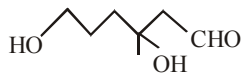


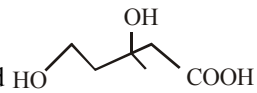
[A]

(1) CH_3COOH and 

(2) CH_3CHO and 

(3) CH_3COCH_2COOH and



(4) CH_3CHO and 

34. N_2 and O_2 are converted to mono cations N_2^+ and O_2^+ respectively, which of the following is wrong?

(1) In N_2^+, the N – N bond weakens

(2) In O_2^+, the O – O bond order increases

(3) In O_2^+, paramagnetism decreases

(4) N_2^+ becomes diamagnetic

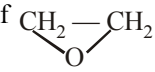
35. The reaction in which hydrogen peroxide acts as a reducing agent is

(1) $PbS + 4H_2O_2 \rightarrow PbSO_4 + 4H_2O$

(2) $2KI + H_2O_2 \rightarrow 2KOH + I_2$

(3) $2FeSO_4 + H_2SO_4 + H_2O_2 \rightarrow Fe_2(SO_4)_3 + 2H_2O$

(4) $Ag_2O + H_2O_2 \rightarrow 2Ag + H_2O + O_2$

36. Reaction of  with RMgX leads to formation of

(1) RCHOHR (2) $RCHOHCH_3$

(3) RCH_2CH_2OH (4) 

37. If *m* and *e* are the mass and charge of the revolving electron in the orbit of radius *r* for hydrogen atom, the total energy of the revolving electron will be:

(1) $\frac{1}{2}\frac{e^2}{r}$ (2) $-\frac{e^2}{r}$

(3) $\frac{me^2}{r}$ (4) $-\frac{1}{2}\frac{e^2}{r}$

38. When chlorine water is added to an aqueous solution of sodium iodide in the presence of chloroform, a violet colouration is obtained. On adding more of chlorine water and vigorous shaking, the violet colour disappears. This shows the conversion of ..... into .....

(1) I_2, HIO_3 (2) I_2, HI

(3) HI, HiO_3 (4) I_2, HOI

39. When tert-butyl chloride is made to react with sodium ethoxide, the major product is

(1) dimethyl ether

(2) di-tert-butyl ether

(3) tert-butylmethyl ether

(4) isobutylene

40. Which statement is true regarding following reactions

$$trans-2-\text{Butene} \xrightarrow{HCO_3H} A$$

$$cis-2-\text{Butene} \xrightarrow{HCO_3H} B$$

(1) Compounds A and B are formed by *syn* addition and they are racemic mixture and meso respectively

(2) Compounds A and B are formed by *anti* addition and are racemic mixture and meso respectively

(3) Compounds A and B are formed by *anti* addition and are meso and racemic mixture respectively

(4) Compounds A and B are formed by *syn* addition and are meso and racemic mixture respectively

41. An organic compound is treated with $NaNO_2$ and dil. HCl at 0°C. The resulting solution is added to an alkaline solution of β-naphthol where by a brilliant red dye is produced. It shows the presence of

(1) $-NO_2$ group

(2) aromatic $-NH_2$ group

(3) $-CONH_2$ group

(4) aliphatic $-NH_2$ group

42. Point out the incorrect statment among the following :

(1) The oxidation state of oxygen is +2 in OF_2.

(2) Acidic character follows the order $H_2O < H_2S < H_2Se < H_2Te$.

(3) The tendency to form multiple bonds increases in moving down the group from sulphur to tellurium (towards C and N).

(4) Sulphur has a strong tendency to catenate while oxygen shows this tendency to a limited extent.

43. Removal of Fe, Cu, W from Sn metal after smelting is by ..... because .....

(1) poling; of more affinity towards oxygen for impurities

(2) selective oxidation; of more affinity towards oxygen for impurities

(3) electrolytic refining; impurities undissolved in elec-trolyte

(4) liquation; Sn having low melting point compared to impurities.

44. Among KO_2, AlO_2^-, BaO_2 and NO_2^+, unpaired electron is present in

(1) NO_2^+ and BaO_2 (2) KO_2 and AlO_2^-

(3) KO_2 only (4) BaO_2 only

45. If a 0.1 M solution of glucose (Mol. wt 180) and 0.1 molar solution of urea (Mol. wt. 60) are placed on two sided semipermeable membrane to equal heights, then it will be correct to say that

(1) there will be no net movement across the membrane.

(2) glucose will flow across the membrane into urea solution.

(3) urea will flow across the membrane into glucose solution.

(4) water will flow from urea solution to glucose solution.

46. Melamine plastic crockery is a codensation polymer of
(1) HCHO and melamine
(2) HCHO and ethylene
(3) melamine and ethylene
(4) None of these

47. When pink complex, $[Co(H_2O)_6]^{2+}$ is dehydrated the colour changes to blue. The correct explanation for the change is :
(1) The octahedral complex becomes square planar.
(2) A tetrahedral complex is formed.
(3) Distorted octahedral structure is obtained.
(4) Dehydration results in the formation of polymeric species.

48. In the form of dichromate, Cr (VI) is a strong oxidising agent in acidic medium but Mo (VI) in MoO_3 and W (VI) in WO_3 are not because _____ .
(i) Cr (VI) is more stable than Mo(VI) and W (VI).
(ii) Mo (VI) and W(VI) are more stable than Cr(VI).
(iii) Higher oxidation states of heavier members of group-6 of transition series are more stable.
(iv) Lower oxidation states of heavier members of group-6 of transition series are more stable.
(1) (i) and (ii) (2) (ii) and (iii)
(3) (i) and (iv) (4) (ii) and (iv)

49. The correct statement among the following is :
(1) The alkali metals when strongly heated in oxygen form superoxides.
(2) Caesium is used in photoelectric cells.
(3) $NaHCO_3$ is more soluble in water than $KHCO_3$.
(4) The size of hydrated ions of alkali metals increases from top to bottom.

50. If the following half cells have E° values as
$A^{3+} + e^- \longrightarrow A^{2+}$, $E° = y_2$ V
$A^{2+} + 2e^- \longrightarrow A$, $E° = -y_1$ V
The E° of the half cell
$A^{3+} + 3e^- \longrightarrow A$ will be
(1) $\frac{2y_1 - y_2}{3}$ (2) $\frac{y_2 - 2y_1}{3}$
(3) $2y_1 - 3y_2$ (4) $y_2 - 2y_1$

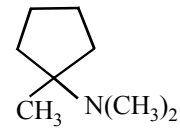
Section - B

51. The radii of Na^+ and Cl^- ions are 100 pm and 200 pm respectively. Calculate the edge length of NaCl unit cell.

52. When 1.8 g of steam at the normal boiling point of water is converted into water at the same temperature, calculate Δs for the above change in J/K. (ΔH_{vap} for water = 40.8 kJ mol^{-1})

53. Given
$n = 5, m_\ell = +1$
Find the maximum number of electron(s) in an atom that can have the quantum numbers as given.

54. What amount of sugar ($C_{12}H_{22}O_{11}$) (in g) is required to prepare 2 L of its 0.1 M aqueous solution?

55. 

Find the number of unsaturated carbon in the product.

56. A tetrapeptide has —COOH group on alanine. This produces glycine (Gly), valine (Val), phenyl alanine (Phe) and alanine (Ala), on complete hydrolysis. For this tetrapeptide, find the number of possible sequences (primary structures) with $—NH_2$ group attached to a chiral center.

57. 4 g of H_2 effused through a pinhole in 10 s at some constant temperature and pressure. Calculate the amount of oxygen effused in the same time interval and at the same conditions of temperature and pressure.

58. The vapour pressure of benzene at a certain temperature is 640 mm of Hg. A non volatile and non electrolyte solid weighing 2.175 g is added to 39.08 g of benzene. If the vapour pressure of the solution is 600 mm of Hg, calculate the molecular weight of solid substance.

59. When CO_2 dissolves in water, the following equilibrium is established

$CO_2 + 2H_2O \rightleftharpoons H_3O^+ + HCO_3^-$; for which the equilibrium constant is 3.8×10^{-6} and pH = 6.0. What would be the ratio of concentration of bicarbonate ion to carbon dioxide?

60. For a first order reaction, calculate the ratio between the time taken to complete 3/4th of the reaction and time taken to complete half of the reaction.

MATHEMATICS

Section - A

61. The roots of the given equation $(p-q)x^2 + (q-r)x + (r-p) = 0$ are :

(1) $\frac{p-q}{r-p}, 1$ (2) $\frac{q-r}{p-q}, 1$

(3) $\frac{r-p}{p-q}, 1$ (4) None of these

62. If a, b, c, d and p are distinct non zero real numbers such that $(a^2 + b^2 + c^2)\,p^2 - 2(ab + bc + cd)p + (b^2 + c^2 + d^2) \leq 0$ then a,b,c,d are in

(1) A.P. (2) GP.

(3) H.P. (4) satisfy ab = cd

63. Which of the following is correct?

(1) If $a^2 + 4b^2 = 12ab$, then

$$\log(a + 2b) = \frac{1}{2}(\log a + \log b)$$

(2) If $\frac{\log x}{b-c} = \frac{\log y}{c-a} = \frac{\log z}{a-b}$,

then $x^a . y^b . z^c = abc$

(3) $\frac{1}{\log_{xy} xyz} + \frac{1}{\log_{yz} xyz} + \frac{1}{\log_{zx} xyz} = 2$

(4) All are correct

64. If $0 < \alpha, \beta, \gamma < \pi/2$ such that $\alpha + \beta + \gamma = \frac{\pi}{2}$ and cot α, cot β, cot γ are in arithmetic progression, then the value of cot α cot γ is

(1) 1 (2) 3

(3) $\cot^2\beta$ (4) cot α + cot γ

65. The equation of the chord of the hyperbola $25x^2 - 16y^2 = 400$ that is bisected at point (5, 3) is:

(1) $135x - 48y = 481$ (2) $125x - 48y = 481$

(3) $125x - 4y = 48$ (4) None of these

66. The number of solutions of the equation

$$\sin\left(\frac{\pi x}{2\sqrt{3}}\right) = x^2 - 2\sqrt{3}x + 4$$

(1) forms an empty set (2) is only one
(3) is only two (4) is more than 2

67. If $1, \alpha_1, \alpha_2, \alpha_3, \alpha_4$ are the roots of $z^5 - 1 = 0$, then the value of $\frac{\omega-\alpha_1}{\omega^2-\alpha_1} \cdot \frac{\omega-\alpha_2}{\omega^2-\alpha_2} \cdot \frac{\omega-\alpha_3}{\omega^2-\alpha_3} \cdot \frac{\omega-\alpha_4}{\omega^2-\alpha_4}$ is (where ω is imaginary cube root of unity)

(1) 1 (2) ω
(3) ω^2 (4) None of these

68. Which of the following is correct?

(1) If A and B are square matrices of order 3 such that $|A| = -1$, $|B| = 3$, then the determinant of 3AB is equal to 27.
(2) If A is an invertible matrix, then $\det(A^{-1})$ is equal to $\det(A)$
(3) If A and B are matrices of the same order, then $(A+B)^2 = A^2 + 2AB + B^2$ is possible if $AB = I$
(4) None of these

69. Let $f(x) = |x - 1|$. Then

(1) $f(x^2) = (f(x))^2$
(2) $f(x+y) = f(x) + f(y)$
(3) $f(|x|) = |f(x)|$
(4) None of these

70. If $\sin^{-1}\frac{2a}{1+a^2} + \sin^{-1}\frac{2b}{1+b^2} = 2\tan^{-1}x$, then x is equal to

(1) $\frac{a-b}{1+ab}$ (2) $\frac{b}{1+ab}$
(3) $\frac{b}{1-ab}$ (4) $\frac{a+b}{1-ab}$

71. Let $f(x) = \frac{2}{x+1}$, $g(x) = \cos x$ and $h(x) = \sqrt{x+3}$ then the range of the composite function fogoh, is

(1) R^+ (2) $R - \{0\}$
(3) $[1, \infty)$ (4) $R^+ - \{1\}$

72. The set of points where $f(x) = (x-1)^2 (x + |x-1|)$ is thrice differentiable, is

(1) R (2) $R - \{0\}$
(3) $R - \{1\}$ (4) $R - \{0,1\}$

73. Let $f(x) = 1/(x-1)$ and $g(x) = 1/(x^2+x-2)$. Then the set of points where (gof)(x) is discontinuous, is

(1) $\{1\}$ (2) $\{-2,1\}$
(3) $\{1/2, 1, 2\}$ (4) $\{1/2, 1\}$

74. $\sum_{r=0}^{m} {}^{n+r}C_n$ is equal to :

(1) ${}^{n+m+1}C_{n+1}$ (2) ${}^{n+m+2}C_n$
(3) ${}^{n+m+3}C_{n-1}$ (4) None of these

75. The value of $\int_{1/n}^{(an-1)/n} \frac{\sqrt{x}}{\sqrt{a-x}+\sqrt{x}}dx$ is

(1) $\frac{a}{2}$ (2) $\frac{1}{2n}(na+2)$

(3) $\frac{na-2}{2n}$ (4) None of these.

76. The value of

$\int_{-\pi/4}^{\pi/4} (x|x|+\sin^3 x + x\tan^2 x + 1)\,dx$ is

(1) 0 (2) 1

(3) $\pi/4$ (4) $\pi/2$

77. Let $(1-x-2x^2)^6 = 1+a_1x+a_2x^2+\ldots.+a_{12}x^{12}$.

Then $\frac{a_2}{2^2}+\frac{a_4}{2^4}+\frac{a_6}{2^6}+.....+\frac{a_{12}}{2^{12}}$ is equal to

(1) –1 (2) –1/2

(3) 0 (4) 1/2

78. The equation of a common tangent to $y^2 = 4x$ and the curve $x^2 + 4y^2 = 8$ can be

(1) $x-2y+2=0$ (2) $x+2y+4=0$

(3) $x-2y=4$ (4) $x+2y=4$

79. The function $f(x) = (x-3)^2$ satisfies all the conditions of mean value theorem in [3, 4]. A point on $y=(x-3)^2$, where the tangent is parallel to the chord joining (3, 0) and (4,1) is

(1) $\left(\frac{7}{2},\frac{1}{2}\right)$ (2) $\left(\frac{7}{2},\frac{1}{4}\right)$

(3) (1,4) (4) (4, 1)

80. If $x+y-z+xyz=0$, then $\frac{2x}{1-x^2}+\frac{2y}{1-y^2}-\frac{2z}{1-z^2}$ is equal to

(1) $\frac{xyz}{[(1-x^2)(1-y^2)(1-z^2)]}$

(2) $\frac{-xyz}{[(1-x^2)(1-y^2)(1-z^2)]}$

(3) $\frac{8xyz}{[(1-x^2)(1-y^2)(1-z^2)]}$

(4) $\frac{-8xyz}{[(1-x^2)(1-y^2)(1-z^2)]}$

Section - B

81. If the number of 5-element subsets of the set $A=\{a_1, a_2, ..., a_{20}\}$ of 20 distinct elements is k times the number of 5-element subsets containing a_4, then k is _____.

82. The value of $\cos 36^0 \cos 42^0 \cos 78^0$ is _____.

$\left[\text{Given : } \sin 18° = \frac{\sqrt{5}-1}{4} \text{ and } \cos 36° = \frac{\sqrt{5}+1}{4}\right]$

83. If $x = 1/5$, the value of $|\cos(\cos^{-1}x + 2\sin^{-1}x)|$ is _____.

84. Let A be the centre of the circle $x^2+y^2-2x-4y-20=0$, and $B(1,7)$ and $D(4,-2)$ are points on the circle then, if tangents be drawn at B and D, which meet at C, then area of quadrilateral $ABCD$ is _____.

85. If the system of linear equations :

$x_1 + 2x_2 + 3x_3 = 6$

$x_1 + 3x_2 + 5x_3 = 9$

$2x_1 + 5x_2 + ax_3 = b$

is consistent and has infinite number of solutions, then value of a + b is _____.

86. If $A^k = 0$ (A is nilpotent with index k), $(I - A)^p = I + A + A^2 + \ldots + A^{k-1}$, then value of $|p|$ is _____.

87. Let $f(x) = \dfrac{x - \{x+1\}}{x - \{x+2\}}$; where {x} is the fractional part of x, then value of $\lim\limits_{x \to 1/3} f(x)$ is _____.

88. Let $y(x)$ be a solution of $\dfrac{(2+\sin x)}{(1+y)}\dfrac{dy}{dx} = \cos x$.

If $y(0) = 2$, then value of $y\left(\dfrac{\pi}{2}\right)$ is _____.

89. $\int \dfrac{1}{f(x)}\,dx = \log(f(x))^2 + C$. If $f(0) = 5$ then value of constant term of f(x) is _____.

90. In a ΔABC, if $\begin{vmatrix} 1 & a & b \\ 1 & c & a \\ 1 & b & c \end{vmatrix} = 0$, then value of $\sin^2 A + \sin^2 B + \sin^2 C$ is _____.